

SWINBURNE UNIVERSITY OF TECHNOLOGY

LOW-COST OPTION

Unit Code: Computing Technology Project A (COS40005)
Project: Internal Assistant for Enterprise Knowledge (Slack chatbot with RAG)
Client: Katalon, Inc. (replacing the original VIB client)
Team: Group 7
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A. Purpose and Scope

This document describes in detail the **Option 2: Low-Cost, Closer-to-Production Environment** for the Katalon internal-knowledge Slack assistant built by Group 7. It explains every technology in the environment, why it was chosen, how it helps the project, and how the parts work together, using an architecture diagram and three sequence diagrams. It complements the *Environment Options* overview (Group 7, 2026a) and the matching project budget workbook (Group 7, 2026b).

The environment keeps the architecture requested by the client: Confluence is the only knowledge source, Slack is the only user interface, and Amazon Bedrock provides retrieval-augmented generation (RAG). It is a **test (staging) environment**. Katalon's production Slack, Confluence and identity provider are not used, because an external student team is unlikely to be given access to them. All prices were checked on 11 November 2026, are in US dollars for the US East (N. Virginia) Region, and may change.

B. Environment at a Glance

Table 1 summarises the environment. Figure 1 shows how the components are connected.

Layer	Technology	Role in the project	Cost to the team
Identity provider	Okta Integrator Free Plan	Single sign-on (SAML) and SCIM provisioning for test personas	US$0
Chat interface	Slack Developer Sandbox + Slack app	Where employees ask questions and receive answers	US$0
Knowledge source	Confluence Standard (Classroom licence)	Sample spaces that copy Katalon's structure, with real permissions	About US$16 per month
Compute	AWS Lambda (bot and sync functions) with a function URL	Runs all logic without servers	Always-free allowance
Caches and state	Amazon DynamoDB with Time to Live (TTL)	Identity map and permission cache	Always-free allowance
Retrieval	Amazon Bedrock Knowledge Bases + Titan Text Embeddings V2	Chunking, embedding and filtered retrieval	Paid from AWS credits
Vector store	Amazon S3 Vectors	Stores embeddings with page-level metadata	Paid from AWS credits
Generation	Amazon Nova Lite on Bedrock (Nova 2 Lite if accuracy needs it)	Writes grounded answers from permitted chunks	Paid from AWS credits
Secrets	AWS Secrets Manager	Holds the Slack and Confluence tokens	US$0.40 per secret per month
Scheduling and monitoring	Amazon EventBridge Scheduler, Amazon CloudWatch, AWS Budgets	Runs the sync, records logs and freshness, alerts on spending	Free allowances

Table 1. Components of the environment

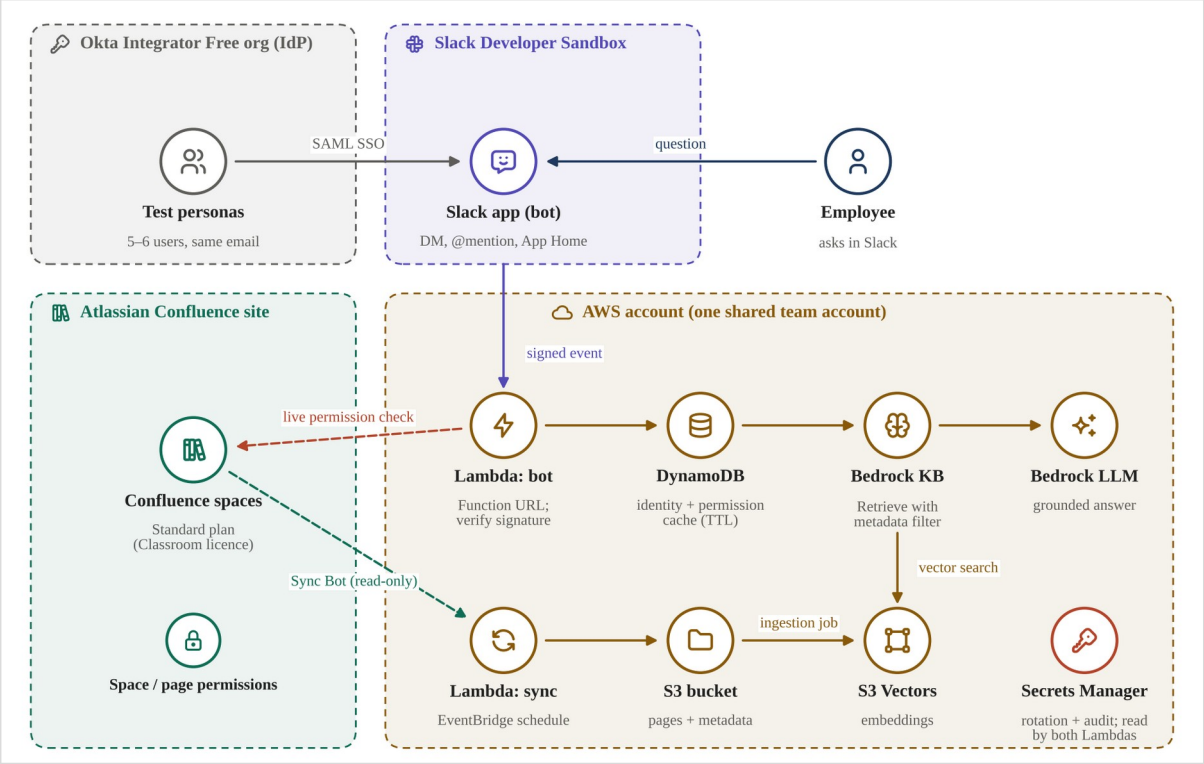


Figure 1. Architecture of Option 2: icons show each service; dashed arrows are permission traffic

C. Components in Detail

Each component below is described with three questions: **what it is**, **why the team uses it**, and **how it helps the project**. The evaluation weights referred to are the client's: accuracy 40%, cost and maintainability 30%, security 20% and creativity 10%.

3.1 Okta Integrator Free Plan (identity provider)

Question	Answer
What it is	A free Okta organisation intended for building and testing integrations. It supports SAML and OIDC single sign-on and SCIM provisioning for custom app integrations (Okta, n.d.).
Why we use it	The Slack sandbox is an Enterprise Grid environment, and Grid requires all users to sign in with SSO, so an identity provider is mandatory (Slack, n.d.-h). Okta offers this at no cost.
How it helps the project	The sign-in path becomes the same as at Katalon: one company login, then Slack. SCIM creates the Slack test users with the same email addresses that the bot later uses to match people to Confluence accounts.

3.2 Slack Developer Sandbox and the Slack app

Question	Answer
What it is	A free, isolated Enterprise Grid sandbox from the Slack Developer Program, limited to eight users and two guests and active for six months by default (Slack, n.d.-a). The bot is a Slack app installed with a bot token (`xoxb`) and the scopes `chat:write`, `app_mentions:read`, `im:history`, `users:read` and `users:read.email`.
Why we use it	It gives the team a real enterprise Slack without touching Katalon's workspace. Eight users are enough for the test personas, and the two guest seats let the team prove that guests are refused.
How it helps the project	Every event carries the sender's Slack ID. The bot calls `users.info` to read the email address and the guest flags `is_restricted` and `is_ultra_restricted` (Slack, n.d.-c, n.d.-d, n.d.-e). Requests are authenticated with the signing secret: Slack signs each request with HMAC-SHA256, and the app compares the signature and rejects timestamps older than five minutes (Slack, n.d.-f).

3.3 Confluence Standard plan (Classroom licence)

Question	Answer
What it is	A Confluence Cloud site on the Standard plan for the whole project, paid through an Atlassian Classroom licence, which gives students cloud subscriptions at 75% off list price (Atlassian, n.d.-b).
Why we use it	Real space and page permissions are available from the first day, so access control can be tested continuously instead of in a single two-week window.
How it helps the project	The site holds sample spaces that copy Katalon's structure (Intranet, Deal Desk, Partner Program and True Platform) and the group names the client shared. A read-only **Sync Bot** account with an API token reads pages and checks permissions; it never edits content (Atlassian, n.d.-e).

3.4 AWS Lambda with a function URL

Question	Answer
What it is	Serverless functions: a **bot function** that handles Slack events and a **sync function** that keeps the knowledge base current. The bot function is reached through a Lambda function URL, a dedicated HTTPS endpoint (Amazon Web Services [AWS], n.d.-f).
Why we use it	There are no servers to patch, and Lambda's free tier includes one million requests per month (AWS, n.d.-g). A function URL removes the need for Amazon API Gateway, whose free calls are not part of the AWS Free plan (AWS Builder Center, 2026).
How it helps the project	Slack expects an HTTP 2xx response within three seconds (Slack, n.d.-g). The bot function verifies the signature, acknowledges at once and continues the work asynchronously, which keeps the user experience responsive.

3.5 Amazon DynamoDB with Time to Live

Question	Answer
What it is	A serverless key-value database holding two caches: the **identity map** (Slack ID to email to Atlassian account ID) and the **permission cache** (allowed page IDs for each user).
Why we use it	Calling Slack and Confluence for every question would be slow and could hit rate limits. TTL deletes expired items automatically and without consuming write throughput (AWS, n.d.-h), and the free tier covers 25 GB of storage (AWS, n.d.-i).
How it helps the project	Answers are faster and cheaper (cost criterion). A 15-minute TTL means that permissions removed in Confluence stop being used quickly, and the live check in Section D removes even that delay for the pages actually used.

3.6 Amazon Bedrock Knowledge Bases with Titan Text Embeddings V2

Question	Answer
What it is	A managed RAG service that reads documents from an S3 bucket, splits them into chunks, embeds them and answers `Retrieve` calls. Titan Text Embeddings V2 produces 1,024-dimension vectors and accepts up to 8,192 tokens per input (AWS, n.d.-j).
Why we use it	It removes the need to write chunking and embedding code. Each page is stored with a `.metadata.json` file containing its page ID, space and groups, and `Retrieve` can filter on these attributes (AWS, 2024, n.d.-k).
How it helps the project	Retrieval only searches pages the asker may read (security), and only changed pages are re-processed during a sync (freshness and cost). Embedding costs about US$0.02 per million tokens (AWS, n.d.-a).

3.7 Amazon S3 Vectors (vector store)

Question	Answer
What it is	Vector storage built into Amazon S3, generally available since December 2025 and supported as a Bedrock knowledge-base vector store (AWS, 2025b).
Why we use it	The default quick-create store, Amazon OpenSearch Serverless, keeps a minimum of four capacity units running and costs roughly US$700 per month even with no queries (Mallesh, n.d.). S3 Vectors has no minimum charge and costs nothing while idle (amitrix, n.d.).
How it helps the project	For a corpus of a few hundred pages, storage and queries cost cents per month, which protects the AWS credits (AWS, n.d.-d). This is the single most important cost decision in the environment.

3.8 Amazon Nova Lite on Bedrock (answer generation)

Question	Answer
What it is	A low-cost foundation model on Bedrock, priced at about US$0.06 per million input tokens and US$0.24 per million output tokens; Nova 2 Lite costs about US$0.30 and US$2.50 (AWS, n.d.-a).
Why we use it	Answers are generated only from retrieved, permitted chunks, so a small model is usually enough. Starting with Nova Lite keeps evaluation runs cheap; the team can switch to Nova 2 Lite if accuracy tests show a need.
How it helps the project	Supports the 40% accuracy criterion through grounded answers with source links, while thousands of test questions still cost only a few dollars.

3.9 AWS Secrets Manager

Question	Answer
What it is	A managed secrets service priced at US$0.40 per secret per month plus US$0.05 per 10,000 API calls (AWS, n.d.-b). It supports automatic rotation and records every access in AWS CloudTrail.
Why we use it	Rotation and audit are standard enterprise practice for credentials, and storing all three tokens in one secret keeps the cost at about US$0.40 per month.
How it helps the project	Holds the Slack bot token, the Slack signing secret and the Sync Bot API token. Both Lambda functions read them at start-up and keep them in memory, so the number of calls stays very low (security criterion).

3.10 EventBridge Scheduler, CloudWatch and AWS Budgets

Question	Answer
What it is	EventBridge Scheduler starts the sync function on a fixed schedule (AWS, n.d.-l). CloudWatch stores logs and custom metrics. AWS Budgets sends an alert when spending passes a threshold (AWS, n.d.-m).
Why we use it	Freshness depends on a reliable schedule, and the team needs evidence (logs, freshness metric) for the evaluation. A budget alert is the last line of defence against unexpected charges.
How it helps the project	Supports the freshness and cost criteria and provides the audit log of who asked what and which pages were used.

D. Workflows and Sequence Diagrams

4.1 Provisioning test personas and signing in

Figure 2 shows how the System Admin (Group 7) prepares a test persona and how the persona signs in. In production, Katalon IT performs the first four steps through its own identity provider; in this environment the team performs them in Okta and Confluence.

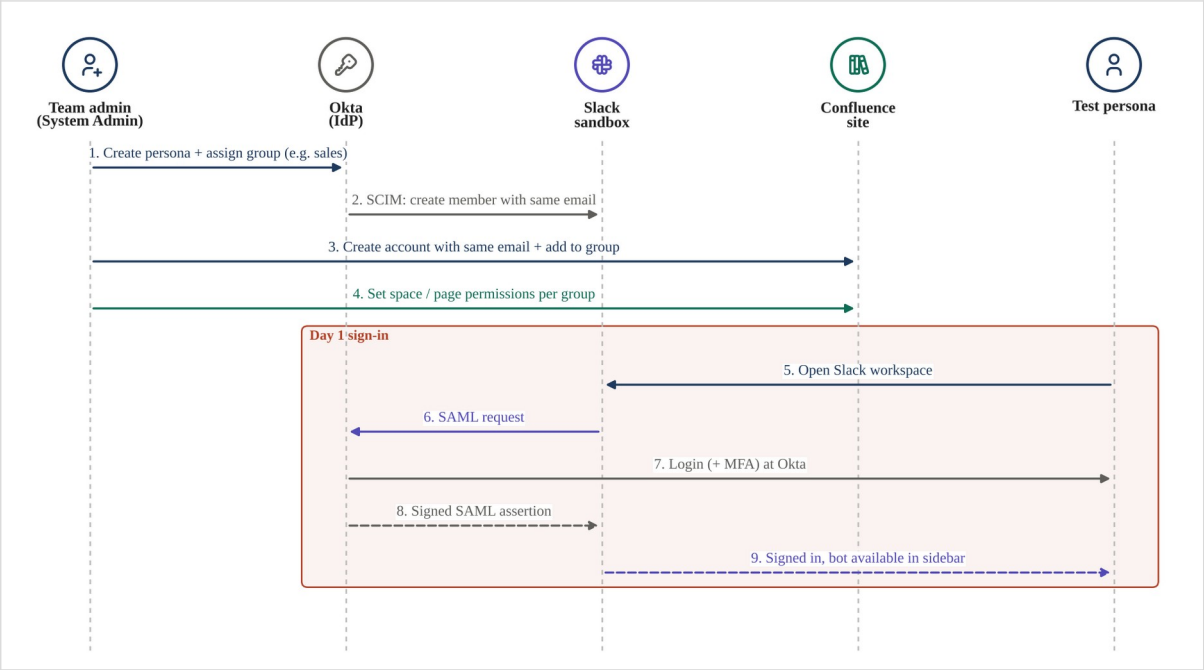


Figure 2. Sequence diagram: provisioning a persona and day-one sign-in

- **Steps 1–2:** the persona is created once in Okta, and SCIM creates the Slack member with the same email (Okta, n.d.).
- **Steps 3–4:** the Confluence account is created manually with the same email, because SCIM into Atlassian needs Atlassian Guard, which is not free for students. The account is added to a group, and the group decides which spaces and pages it can read.
- **Steps 5–9:** SAML single sign-on (Slack, n.d.-b). The bot never sees a password; it only receives the persona's Slack ID with each event.

4.2 Answering a question with two-layer access control

Figure 3 is the core workflow. The asker never signs in to the bot separately; identity comes from Slack, and access comes from Confluence.

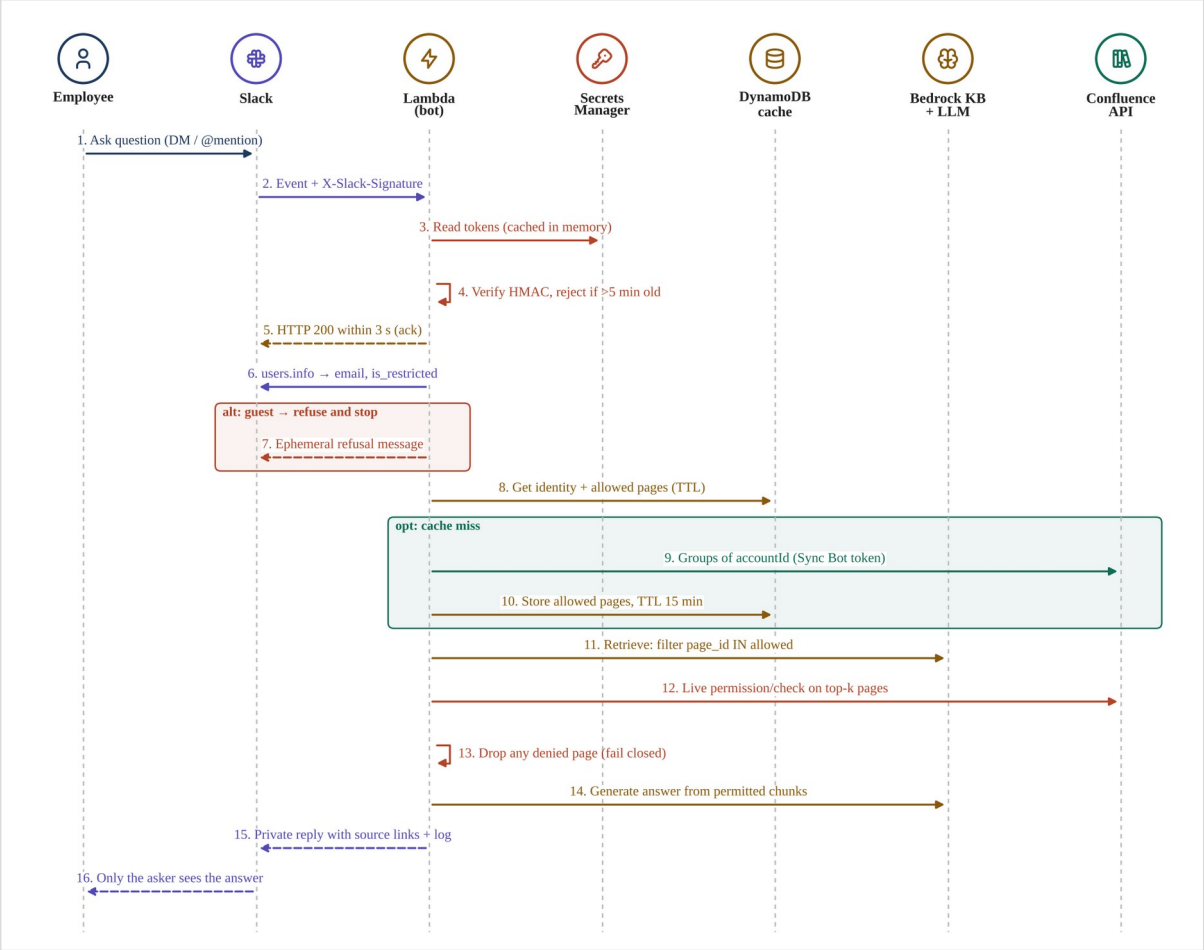


Figure 3. Sequence diagram: answering a question with a pre-filter and a live permission check

1. **Authenticate the request (steps 2–5).** The bot reads its tokens from Secrets Manager, verifies the Slack signature and acknowledges within three seconds (Slack, n.d.-f, n.d.-g).
2. **Identify the asker (steps 6–7).** `users.info` returns the email and guest flags; guests receive a private refusal and processing stops (Slack, n.d.-c).
3. **Resolve permissions (steps 8–10).** The allowed page IDs come from the DynamoDB cache, or from Confluence on a cache miss, and are cached with a 15-minute TTL.
4. **Layer 1, pre-filter (step 11).** Bedrock `Retrieve` searches only chunks whose `page_id` is in the allowed list (AWS, n.d.-k).
5. **Layer 2, live check (steps 12–13).** The top results are checked again with the Confluence permission-check API (Atlassian, n.d.-d). Any page that is denied, or cannot be checked, is dropped: the system fails closed (NHI Mgmt Group, n.d.).
6. **Answer privately (steps 14–16).** The model writes the answer only from permitted chunks, and the reply is visible only to the asker, with source links. The question, user and page IDs are logged.

The second layer exists because a metadata filter is only as reliable as the identity and metadata passed to it. Recent RAG security research shows that checking identity only before retrieval is not sufficient, and recommends access decisions at both the retrieval and the generation stage (“Towards Secure Retrieval-Augmented Generation,” 2026; Yesil et al., 2026).

4.3 Keeping the knowledge base fresh

Figure 4 shows the incremental sync that serves the freshness criterion.

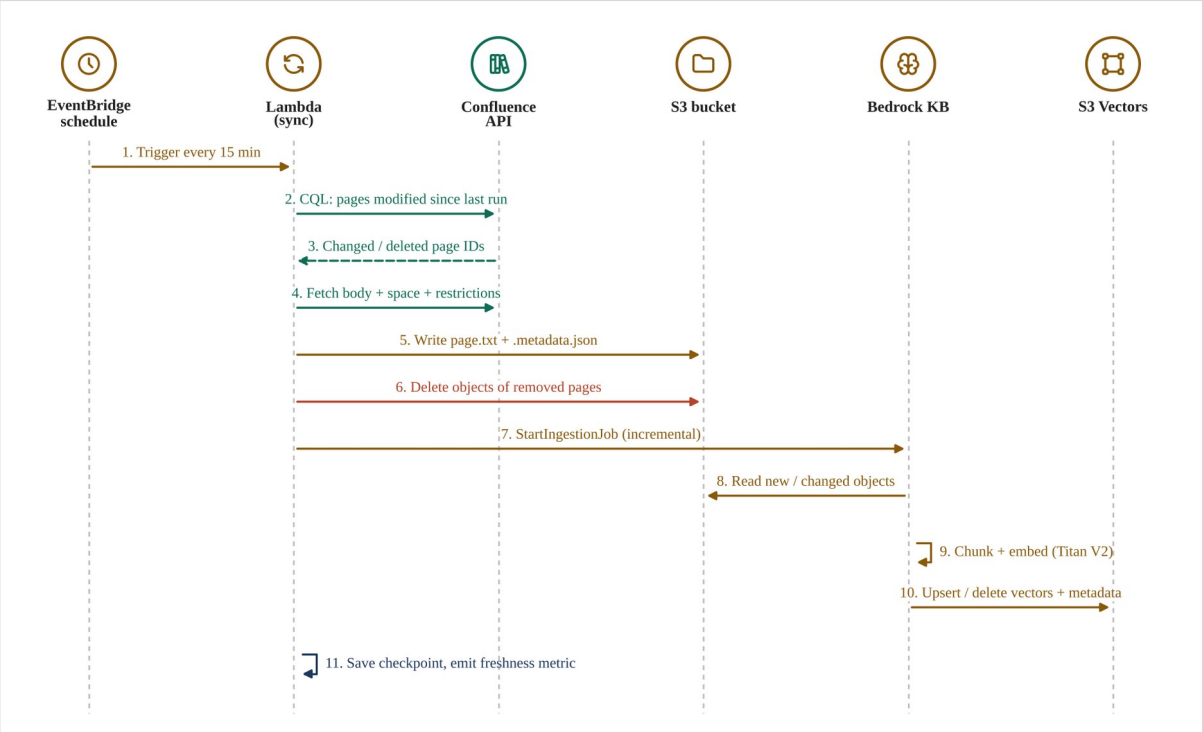


Figure 4. Sequence diagram: incremental synchronisation from Confluence to Bedrock

- **Change detection:** every 15 minutes the sync function asks Confluence, using CQL, for pages modified since the last checkpoint (Atlassian, n.d.-f).
- **Metadata:** each page is written to S3 with a `.metadata.json` file (page ID, space, groups), which is what makes the permission filter in Figure 3 possible (AWS, n.d.-k).
- **Incremental ingestion:** a Bedrock ingestion job processes only new, changed or deleted objects, so cost stays proportional to change (AWS, n.d.-n).
- **Evidence:** the function stores a checkpoint and publishes a freshness metric (minutes since the last successful sync) to CloudWatch for the evaluation.

E. Testing in This Environment

5.1 Test personas

Persona	Slack account	Confluence group	Expected result
Employee – Sales	Member	deal-desk	Answers only from Deal Desk and public Intranet pages
Employee – Partner	Member	partner-program	Answers only from Partner Program and public Intranet pages
Content Owner	Member	true-platform-admins (space admin)	Answers from True Platform; can force a re-sync
Client Reviewer	Member	reviewers	Same answers as an Employee; read-only dashboard
System Admin	Owner / Admin	none beyond own access	Manages the system but cannot read restricted pages
Guest	Single-channel guest	none	Refused with a private message

Table 2. Test personas (at most eight Slack users and two guests in the sandbox)

5.2 How access control is tested

1. **From the first sprint:** permissions are configured in Confluence and the persona matrix in Table 2 is run automatically after every deployment (regression test).
2. **Change tests:** a page restriction is removed and re-added to measure how quickly the bot reacts (cache TTL plus live check).
3. **UAT:** the Client Reviewer can test whenever convenient, because there is no trial deadline.

F. Timeline

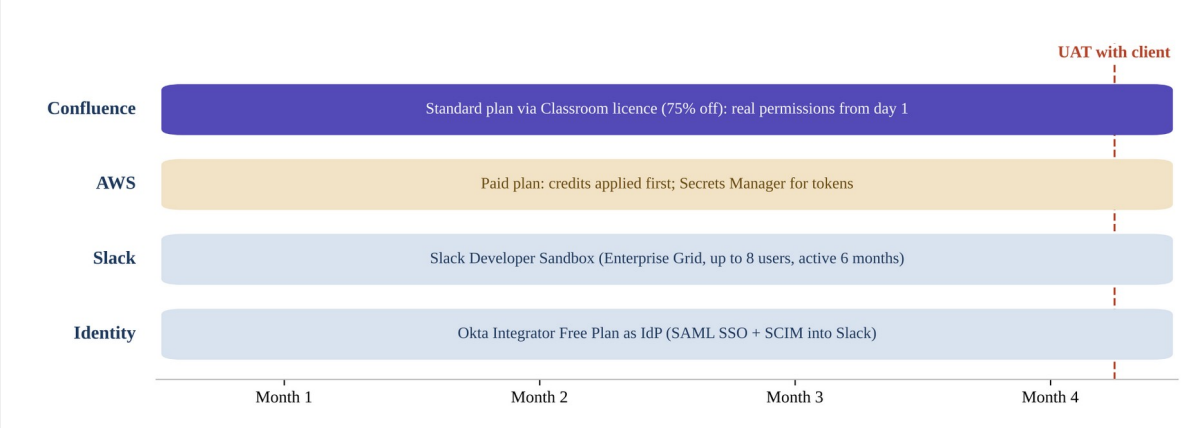


Figure 5. Four-month timeline of Option 2

Nothing in this option has a hard deadline inside the project. The Classroom licence must be applied for early, because Atlassian may grant or deny each application (Atlassian, n.d.-a).

G. Cost Summary

The detailed cost estimate, with labour hours and milestones, is in the matching budget workbook (Group 7, 2026b). The summary is:

Item	Estimate (4 months)	Paid by
Confluence Standard, Classroom licence (75% off)	About US$64	Team
AWS Secrets Manager (one secret)	About US$2	Team
Bedrock, S3 Vectors, Lambda, DynamoDB and other AWS usage	About US$7–15	AWS Free Tier credits applied first
Reserve buffer (about 10%)	About US$7	Team, only if needed
Cash cost to the team	**About US$73, about US$12 per member for six members**	

New AWS customers receive US$100 in credits at sign-up and can earn another US$100 by using services such as Amazon EC2 and Amazon Bedrock (AWS, 2025a). On the Paid plan the credits are applied first and only usage above them is charged (AWS, 2025a); the always-free allowances still apply (AWS, n.d.-e).

H. Risks and Mitigations

Risk	Effect	Mitigation
Classroom licence refused or delayed	No Confluence permissions	Apply in week 1; use the 14-day Standard trial from Option 1 as a stopgap
Card holder leaves or the subscription lapses	Confluence returns to Free and permissions stop working	Name one owner and one backup; add the renewal date to the team schedule
Someone selects OpenSearch Serverless by mistake	About US$700 per month, credits gone in about a week	Use S3 Vectors only; AWS Budgets alert at US$10
Tokens leak through code or logs	Security incident	Store tokens only in Secrets Manager; never log tokens; rotate the Slack token if exposed
Sandbox data differs from Katalon's real content	Accuracy results do not transfer	Build sample pages from the structure the client shared and ask the client to review them

I. Alignment with Evaluation Criteria

Criterion (weight)	How this environment supports it
Accuracy (40%)	Grounded answers from permitted chunks only, with source links; low model cost allows many evaluation runs.
Cost and maintainability (30%)	About US$73 in total; same serverless design; production-like secret rotation reduces later rework.
Security (20%)	Signature verification, guest refusal, two-layer permission check that fails closed, private replies, tokens in Secrets Manager, audit log. Real permissions are tested throughout the project.
Creativity (10%)	Production-like practice (continuous permission regression tests, secret rotation) on a student budget.

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